

3.1

$$\mathcal{L}' = \mathcal{L}(\vec{q}, \dot{\vec{q}}, t) + \frac{dF(\vec{q}, t)}{dt}$$

$F(\vec{q}, t)$ arbitrary function. Prove E-L for $\mathcal{L}'$

$$\frac{dF}{dt} = \sum_k \frac{\partial F}{\partial q_k} \dot{q}_k + \frac{\partial F}{\partial t}$$

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}'}{\partial \dot{q}_i} \right) &= \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) + \frac{d}{dt} \left(\frac{\partial}{\partial \dot{q}_i} \left[\sum_k \frac{\partial F}{\partial q_k} \dot{q}_k + \frac{\partial F}{\partial t} \right] \right) \\ &= \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) + \frac{d}{dt} \left(\frac{\partial F}{\partial q_i} \right) \end{aligned}$$

$$\frac{\partial \mathcal{L}'}{\partial q_i} = \frac{\partial \mathcal{L}}{\partial q_i} + \frac{\partial}{\partial q_i} \left(\frac{dF}{dt} \right)$$

Now, as in a change of coordinates, the order of the derivatives can be exchanged because

$$\frac{\partial}{\partial q_i} \left(\frac{dF}{dt} \right) = \frac{\partial}{\partial q_i} \left(\sum_k \frac{\partial F}{\partial q_k} \dot{q}_k + \frac{\partial F}{\partial t} \right) =$$

$$= \sum_k \frac{\partial}{\partial q_k} \left(\frac{\partial F}{\partial q_i} \right) \dot{q}_k + \frac{\partial}{\partial t} \left(\frac{\partial F}{\partial q_i} \right) = \frac{d}{dt} \left(\frac{\partial F}{\partial q_i} \right)$$

therefore $\frac{d}{dt} \left(\frac{\partial \mathcal{L}'}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}'}{\partial q_i} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) + \frac{d}{dt} \left(\frac{\partial F}{\partial q_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial F}{\partial q_i} \right) = 0$

By E-L

Alternative

$$\begin{aligned} S' &= \int_{t_1}^{t_2} \mathcal{L}' dt = \int_{t_1}^{t_2} dt \mathcal{L} + \int_{t_1}^{t_2} dt \frac{dF}{dt} = \\ &= S + F \Big|_{q_1, t_1}^{q_2, t_2} = S + \text{constant} \end{aligned}$$

Therefore, the trajectory that minimizes S also minimizes S' .

3.2

$$\mathcal{L}(q, \dot{q}, \ddot{q}, t)$$

Prove the "modified" E-L equations

$$\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) + \frac{d^2}{dt^2} \left(\frac{\partial \mathcal{L}}{\partial \ddot{q}} \right) = 0$$

$$\delta S = \int_{t_1}^{t_2} dt \mathcal{L}(q+\eta, \dot{q}+\dot{\eta}, \ddot{q}+\ddot{\eta}) - \int_{t_1}^{t_2} dt \mathcal{L}(q, \dot{q}, \ddot{q}, t)$$

Now:

$$\begin{aligned} \mathcal{L}(q+\eta, \dot{q}+\dot{\eta}, \ddot{q}+\ddot{\eta}, t) &= \mathcal{L}(q, \dot{q}, \ddot{q}, t) + \frac{\partial \mathcal{L}}{\partial q} \eta + \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{\eta} + \\ &+ \frac{\partial \mathcal{L}}{\partial \ddot{q}} \ddot{\eta} + \mathcal{O}(\eta^2) \end{aligned}$$

Therefore

$$\delta S = \int_{t_1}^{t_2} dt \left(\frac{\partial \mathcal{L}}{\partial q} \eta + \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{\eta} + \frac{\partial \mathcal{L}}{\partial \ddot{q}} \ddot{\eta} \right) + \mathcal{O}(\eta^2)$$

and the condition for stationary action is

$$\int_{t_1}^{t_2} dt \left(\frac{\partial \mathcal{L}}{\partial q} \eta + \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{\eta} + \frac{\partial \mathcal{L}}{\partial \ddot{q}} \ddot{\eta} \right) = 0$$

Now we integrate the 2nd and 3rd terms by parts:

$$\int_{t_1}^{t_2} dt \frac{\partial \mathcal{L}}{\partial \dot{q}} \dot{q} = \left[\eta \frac{\partial \mathcal{L}}{\partial \dot{q}} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} dt \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) \eta$$

$u = \frac{\partial \mathcal{L}}{\partial \dot{q}}, v = \eta$
 $0, \eta(t_1) = \eta(t_2) = 0$

$$\int_{t_1}^{t_2} dt \frac{\partial \mathcal{L}}{\partial \ddot{q}} \ddot{q} = \left[\dot{\eta} \frac{\partial \mathcal{L}}{\partial \ddot{q}} \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} dt \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \ddot{q}} \right) \dot{\eta}$$

$u = \frac{\partial \mathcal{L}}{\partial \ddot{q}}, v = \dot{\eta}$
 $0, \dot{\eta}(t_1) = \dot{\eta}(t_2) = 0$

$$= - \left[\eta \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \ddot{q}} \right) \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} dt \frac{d^2}{dt^2} \left(\frac{\partial \mathcal{L}}{\partial \ddot{q}} \right) \eta$$

0

Putting all together, we get

$$\int_{t_1}^{t_2} dt \left[\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) + \frac{d^2}{dt^2} \left(\frac{\partial \mathcal{L}}{\partial \ddot{q}} \right) \right] \eta = 0 \Leftrightarrow$$

$$\Leftrightarrow \frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) + \frac{d^2}{dt^2} \left(\frac{\partial \mathcal{L}}{\partial \ddot{q}} \right) = 0 \quad \text{q.e.d.}$$

Now consider the Lagrangian

$$\mathcal{L} = -\frac{m}{2} \dot{q}^2 - \frac{k}{2} q^2$$

$$\frac{\partial \mathcal{L}}{\partial q} = -\frac{m \ddot{q}}{2} - k q, \quad \frac{\partial \mathcal{L}}{\partial \dot{q}} = 0, \quad \frac{\partial \mathcal{L}}{\partial \ddot{q}} = -\frac{m q}{2}$$

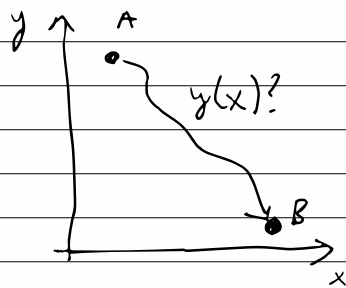
$$\frac{d^2}{dt^2} \left(\frac{\partial \mathcal{L}}{\partial \ddot{q}} \right) = \frac{d^2}{dt^2} \left(-\frac{m q}{2} \right) = -\frac{m \ddot{q}}{2}$$

$$E-L \Rightarrow -\frac{m}{2} \ddot{q} - k q - \frac{m}{2} \ddot{q} = 0 \Rightarrow \boxed{m \ddot{q} = -k q}$$

Harmonic
oscillator

(5.3)

$$S = \int_A^B ds$$



We aim to find the trajectory $y(x)$ that minimizes S .

$$ds = \sqrt{dx^2 + dy^2} = dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = dx \sqrt{1 + \dot{y}^2}$$

$$S = \int_{x_A}^{x_B} dx \sqrt{1 + \dot{y}^2} \Rightarrow \text{Our "Lagrangian" is } \mathcal{L} = \sqrt{1 + \dot{y}^2}$$

Apply E-L equations to $\mathcal{L}(y, \dot{y}, x)$

$$\frac{\partial \mathcal{L}}{\partial y} = 0 \quad ; \quad \frac{\partial \mathcal{L}}{\partial \dot{y}} = \frac{1}{2} (1 + \dot{y}^2)^{-1/2} \cdot 2\dot{y} = \sqrt{\frac{\dot{y}^2}{1 + \dot{y}^2}}$$

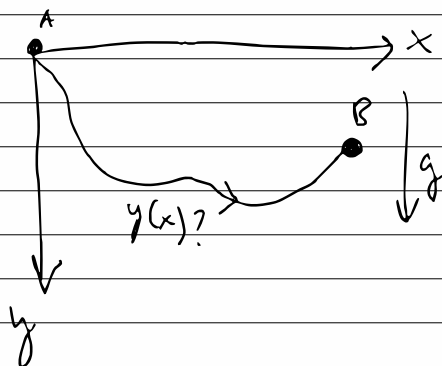
$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) = 0 \quad \Leftrightarrow \quad \sqrt{\frac{\dot{y}^2}{1 + \dot{y}^2}} = C = \text{constant}$$

$$\Rightarrow \frac{\dot{y}^2}{1 + \dot{y}^2} = C^2 \Rightarrow \dot{y}^2 (1 - C^2) = C^2 \Rightarrow \dot{y} = \sqrt{\frac{C^2}{1 - C^2}}$$

$$\Rightarrow \boxed{y = \sqrt{\frac{C^2}{1 - C^2}} x + A} \quad \text{g.c.d.}$$

3.4

Brachistochrone



$y(x)$ such that
time between A and
B is minimal

a) We let v the velocity along the curve. Then

$$T = \int_A^B \frac{ds}{v}$$

Now, with the axes as drawn here, conservation of energy (no friction) implies

$$\frac{m}{2} v^2 = mgy \Rightarrow v = \sqrt{2gy} \quad \text{As in (3.1)}$$

$$\text{Then } T = \int_{x_A}^{x_B} dx \frac{\sqrt{1 + \dot{y}^2}}{\sqrt{2gy}} = \int_{x_A}^{x_B} dx \sqrt{\frac{1 + \dot{y}^2}{2gy}} \Rightarrow$$

$$\Rightarrow \text{our "Lagrangian" is } \mathcal{L} = \sqrt{\frac{1 + \dot{y}^2}{2gy}}$$

no x dependency

$$b) \mathcal{L}(y, \dot{y}, x) = \mathcal{L}(y, \dot{y}) \Rightarrow \dot{y} \frac{\partial \mathcal{L}}{\partial \dot{y}} - \mathcal{L} = \text{constant}$$

(?)

$$\dot{y} \frac{\partial \mathcal{L}}{\partial \dot{y}} - \mathcal{L} = \text{constant} \Leftrightarrow \frac{d}{dt} \left(\dot{y} \frac{\partial \mathcal{L}}{\partial \dot{y}} - \mathcal{L} \right) = 0$$

$$\frac{d}{dt} \left(\dot{y} \frac{\partial \mathcal{L}}{\partial \dot{y}} - \mathcal{L} \right) = \ddot{y} \frac{\partial \mathcal{L}}{\partial \dot{y}} + \dot{y} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{y}} \right) - \frac{d\mathcal{L}}{dt} =$$

E-L

$$= \ddot{y} \frac{\partial \mathcal{L}}{\partial \dot{y}} + \dot{y} \frac{\partial \mathcal{L}}{\partial y} - \left(\frac{\partial \mathcal{L}}{\partial y} \dot{y} + \frac{\partial \mathcal{L}}{\partial \dot{y}} \ddot{y} \right) = 0 \quad \text{qed}$$

$$c) \dot{y} \frac{\partial \mathcal{L}}{\partial \dot{y}} - \mathcal{L} = \dot{y} \frac{1}{x} \left(\frac{1 + \dot{y}^2}{2gy} \right)^{-1/2} \frac{(2\dot{y})}{2gy} - \left(\frac{1 + \dot{y}^2}{2gy} \right)^{1/2} =$$

$$= \frac{\dot{y}^2}{\sqrt{1 + \dot{y}^2} \sqrt{2gy}} - \frac{\sqrt{1 + \dot{y}^2}}{\sqrt{2gy}} = \frac{\dot{y}^2 - 1 - \dot{y}^2}{\sqrt{1 + \dot{y}^2} \sqrt{2gy}} =$$

$$= - \frac{1}{\sqrt{1 + \dot{y}^2} \sqrt{2gy}} \neq \text{ct} \Rightarrow \boxed{y(1 + \dot{y}^2) = A} \quad \square$$

$$d) \quad \boxed{y = \frac{A}{2} (1 - \cos \theta)} \quad \neq A \sin^2 \frac{\theta}{2}$$

$$y(1 + \dot{y}^2) = A \Rightarrow \dot{y}^2 = \frac{A}{y} - 1$$

$$\pm \frac{dy}{dx} = \sqrt{\frac{A-y}{y}} \Rightarrow \sqrt{\frac{y}{A-y}} dy = \pm dx$$

$$dy = \frac{dy}{d\theta} d\theta = 2A \sin \frac{\theta}{2} \cos \frac{\theta}{2} \cdot \frac{1}{2} = A \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta$$

$$\sqrt{\frac{A \sin^2 \frac{\theta}{2}}{A(1 - \sin^2 \frac{\theta}{2})}} A \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta = \pm dx \Rightarrow$$

$$= \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} A \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta = \pm dx \Rightarrow$$

$$= A \sin^2 \frac{\theta}{2} d\theta = \pm dx \Rightarrow \quad \swarrow \sin^2 \frac{\theta}{2} = (1 - \cos \theta)/2$$

$$\Rightarrow \int \frac{A}{2} (1 - \cos \theta) d\theta = \int \pm dx \Rightarrow \boxed{x = \pm \frac{A}{2} (\theta - \sin \theta) + C'}$$

3.5

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta, \quad V(r)\end{aligned}$$

$$\dot{x} = \dot{r} \cos \theta - r \sin \theta \dot{\theta}; \quad \dot{x}^2 = \dot{r}^2 \cos^2 \theta + r^2 \dot{\theta}^2 \sin^2 \theta - 2r\dot{r} \cos \theta \dot{\theta}$$

$$\dot{y} = \dot{r} \sin \theta + r \cos \theta \dot{\theta}; \quad \dot{y}^2 = \dot{r}^2 \sin^2 \theta + r^2 \dot{\theta}^2 \cos^2 \theta + 2r\dot{r} \sin \theta \dot{\theta}$$

$$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2 \dot{\theta}^2$$

$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r)$$

$$\boxed{\theta} \quad \left. \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = \frac{d}{dt} (m r^2 \dot{\theta}) \right\} \stackrel{E-L}{\Rightarrow} \frac{d}{dt} (m r^2 \dot{\theta}) = 0 \Rightarrow \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\Rightarrow \boxed{m r^2 \dot{\theta} = l_0} \quad \downarrow \quad \begin{matrix} \text{constant} \\ \dot{\theta} = \frac{l_0}{m r^2} \end{matrix}$$

$$\boxed{r} \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) = \frac{d}{dt} (m \dot{r}) = m \ddot{r}$$

$$\frac{\partial \mathcal{L}}{\partial r} = m r \dot{\theta}^2 - \frac{dV(r)}{dr} \stackrel{\text{from above}}{=} \frac{m r l_0^2}{m^2 r^4} - \frac{dV}{dr} = \frac{l_0^2}{m r^3} - \frac{dV}{dr}$$

$$\stackrel{E-L}{\Rightarrow} \boxed{m \ddot{r} = \frac{l_0^2}{m r^3} - \frac{dV}{dr}}$$

3.6

$$\begin{aligned}x &= r \sin \theta \cos \phi \\y &= r \sin \theta \sin \phi \\z &= r \cos \theta\end{aligned}$$

$$\begin{aligned}\dot{x} &= \dot{r} \sin \theta \cos \phi + r \dot{\theta} \cos \theta \cos \phi - r \dot{\phi} \sin \theta \sin \phi \\ \dot{y} &= \dot{r} \sin \theta \sin \phi + r \dot{\theta} \cos \theta \sin \phi + r \dot{\phi} \sin \theta \cos \phi \\ \dot{z} &= \dot{r} \cos \theta - r \dot{\theta} \sin \theta\end{aligned}$$

$$\dot{x}^2 = \dot{r}^2 \sin^2 \theta \cos^2 \phi + r^2 \dot{\theta}^2 \cos^2 \theta \cos^2 \phi + r^2 \dot{\phi}^2 \sin^2 \theta \sin^2 \phi + 2r \dot{r} \dot{\theta} \sin \theta \cos \theta \cos^2 \phi - 2r \dot{r} \dot{\phi} \sin^2 \theta \cos \phi \sin \phi - 2r^2 \dot{\theta} \dot{\phi} \sin \theta \cos \theta \sin \phi \cos \phi$$

$$\dot{y}^2 = \dot{r}^2 \sin^2 \theta \sin^2 \phi + r^2 \dot{\theta}^2 \cos^2 \theta \sin^2 \phi + r^2 \dot{\phi}^2 \sin^2 \theta \cos^2 \phi + 2r \dot{r} \dot{\theta} \sin \theta \cos \theta \sin^2 \phi + 2r \dot{r} \dot{\phi} \sin^2 \theta \sin \phi \cos \phi + 2r^2 \dot{\theta} \dot{\phi} \sin \theta \cos \theta \sin \phi \cos \phi$$

$$\dot{z}^2 = \dot{r}^2 \cos^2 \theta + r^2 \dot{\theta}^2 \sin^2 \theta - 2r \dot{r} \dot{\theta} \sin \theta \cos \theta$$

$$T = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2 \sin^2 \theta) \Rightarrow \mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2 \sin^2 \theta) + V(r)$$

Equations of motion

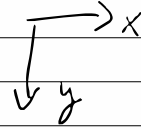
$$\boxed{\phi} \quad \left. \begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) &= \frac{d}{dt} (m r^2 \sin^2 \theta \dot{\phi}) \\ \frac{\partial \mathcal{L}}{\partial \phi} &= 0 \end{aligned} \right\} \Rightarrow \boxed{m r^2 \sin^2 \theta \dot{\phi} = \text{constant}}$$

$$\boxed{\theta} \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = \frac{d}{dt} (m r^2 \dot{\theta}) = 2m r \dot{r} \dot{\theta} + m r^2 \ddot{\theta} \quad \Rightarrow$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = m r^2 \dot{\phi}^2 \sin \theta \cos \theta$$

$$\Rightarrow \boxed{2m r \dot{r} \dot{\theta} + m r^2 \ddot{\theta} - m r^2 \dot{\phi}^2 \sin \theta \cos \theta = 0}$$

3.7



We choose θ ; l as our generalized coordinates

$$\begin{aligned} x &= l \sin \theta; \quad \dot{x} = \dot{l} \sin \theta + l \cos \theta \dot{\theta}; \quad \ddot{x} = \ddot{l} \sin \theta + l^2 \cos^2 \theta \dot{\theta}^2 + 2\dot{l} \cos \theta \dot{\theta} \dots \\ y &= l \cos \theta; \quad \dot{y} = \dot{l} \cos \theta - l \sin \theta \dot{\theta}; \quad \ddot{y} = \ddot{l} \cos \theta + l^2 \sin^2 \theta \dot{\theta}^2 - 2\dot{l} \sin \theta \dot{\theta} \dots \end{aligned}$$

$$\frac{m}{2} (\dot{x}^2 + \dot{y}^2) = \frac{m}{2} (\dot{l}^2 + l^2 \dot{\theta}^2) + 0$$

$$\mathcal{L} = \frac{m}{2} (\dot{l}^2 + l^2 \dot{\theta}^2) - \frac{k}{2} (l - l_0)^2 + mgl \cos \theta$$

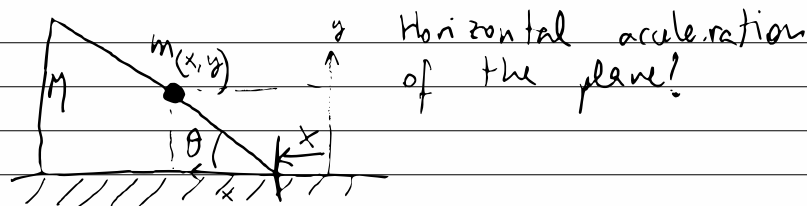
$$\left[\begin{aligned} \frac{\partial \mathcal{L}}{\partial \theta} &= -mgl \sin \theta \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) &= \frac{d}{dt} (ml^2 \dot{\theta}) = 2ml \dot{l} \dot{\theta} + ml^2 \ddot{\theta} \end{aligned} \right] \Rightarrow$$

$$\Rightarrow l^2 \ddot{\theta} + 2l \dot{l} \dot{\theta} + gl \sin \theta = 0 \Rightarrow \boxed{l \ddot{\theta} + 2 \dot{l} \dot{\theta} + g \sin \theta = 0}$$

$$\left[\begin{aligned} \frac{\partial \mathcal{L}}{\partial l} &= m \dot{\theta}^2 - k(l - l_0) + mg \cos \theta \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{l}} \right) &= m \ddot{l} \end{aligned} \right] \Rightarrow$$

$$\Rightarrow \boxed{m \ddot{l} = m \dot{\theta}^2 - k(l - l_0) + mg \cos \theta}$$

3.8



2 degrees of freedom: X of the plane M and x of the block m

$$T = \frac{M}{2} \dot{X}^2 + \frac{m}{2} (\dot{x}^2 + \dot{y}^2) \quad , \quad V = mgy$$

$$\frac{y}{x-X} = \tan \theta \Rightarrow y = (x-X) \tan \theta$$

$$\dot{y} = (\dot{x} - \dot{X}) \tan \theta; \quad \dot{y}^2 = \dot{x}^2 \tan^2 \theta + \dot{X}^2 \tan^2 \theta - 2\dot{x}\dot{X} \tan^2 \theta$$

$$\mathcal{L} = \frac{M}{2} \dot{X}^2 + \frac{m}{2} \left(\dot{x}^2 \left(1 + \frac{\sin^2 \theta}{\cos^2 \theta} \right) + \dot{X}^2 \tan^2 \theta - 2\dot{x}\dot{X} \tan^2 \theta \right)$$

$$- mgy = - mg(x-X) \tan \theta =$$

$$= \frac{\dot{X}^2}{2} (M + m \tan^2 \theta) + \frac{\dot{x}^2 m}{2 \cos^2 \theta} - m\dot{x}\dot{X} \tan^2 \theta - mg(x-X) \tan \theta$$

$$\boxed{\times} \quad \frac{\partial \mathcal{L}}{\partial x} = -mg \tan \theta$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = \frac{d}{dt} \left(\frac{m \dot{x}}{\cos^2 \theta} - m \dot{X} \tan \theta \right) = \frac{m}{\cos^2 \theta} \ddot{x} - m \tan^2 \theta \ddot{X} \quad \left. \vphantom{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right)} \right\} \Rightarrow$$

$$\frac{m}{\cos^2 \theta} \ddot{x} - m \tan^2 \theta \ddot{X} + mg \tan \theta = 0 \quad (\Delta)$$

$$\boxed{\times} \quad \frac{\partial \mathcal{L}}{\partial X} = -mg \tan \theta; \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{X}} \right) = \frac{d}{dt} \left(\dot{X} (M + m \tan^2 \theta) - m \dot{x} \tan \theta \right) = \left. \vphantom{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{X}} \right)} \right\} \Rightarrow$$

$$= \ddot{X} (M + m \tan^2 \theta) - m \ddot{x} \tan \theta$$

$$\Rightarrow \ddot{X} (M + m \tan^2 \theta) - m \ddot{x} \tan \theta - mg \tan \theta = 0 \quad (\square)$$

$$(\Delta) \quad \ddot{x} = \cos^2 \theta (\tan^2 \theta \ddot{X} - g \tan \theta)$$

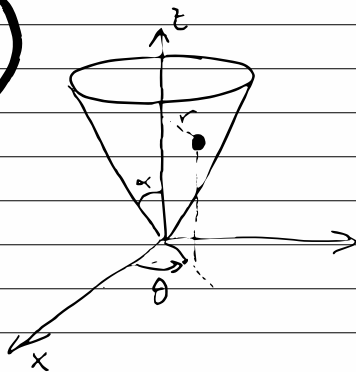
$$(\square) \quad \Rightarrow \ddot{X} (M + m \tan^2 \theta) - m \tan^2 \theta \cos^2 \theta (\tan^2 \theta \ddot{X} - g \tan \theta) - mg \tan \theta = 0$$

$$\Rightarrow \ddot{X} M + \ddot{X} m \tan^2 \theta - \ddot{X} m \sin^2 \theta \tan^2 \theta + mg \tan \theta (\sin^2 \theta - 1) = 0$$

$$\Rightarrow \ddot{X} (M + m \sin^2 \theta) - mg \sin \theta \cos \theta = 0 \Rightarrow$$

$$\Rightarrow \boxed{\ddot{X} = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta}}$$

3.9



Equations of motion for r and θ ?

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\tan \alpha = \frac{r}{z} \Rightarrow z = \frac{r}{\tan \alpha}$$

$$\begin{aligned} \dot{x} &= \dot{r} \cos \theta - r \sin \theta \dot{\theta} ; \dot{x}^2 = \dot{r}^2 \cos^2 \theta + r^2 \dot{\theta}^2 \sin^2 \theta - 2r\dot{r}\sin\theta\cos\theta\dot{\theta} \\ \dot{y} &= \dot{r} \sin \theta + r \cos \theta \dot{\theta} ; \dot{y}^2 = \dot{r}^2 \sin^2 \theta + r^2 \dot{\theta}^2 \cos^2 \theta + 2r\dot{r}\sin\theta\cos\theta\dot{\theta} \\ \dot{z} &= \frac{\dot{r}}{\tan \alpha} ; \dot{z}^2 = \frac{\dot{r}^2}{\tan^2 \alpha} \end{aligned}$$

$$\begin{aligned} \mathcal{L} &= \frac{m}{2} (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz = \frac{m}{2} \left(\dot{r}^2 \left(1 + \frac{1}{\tan^2 \alpha} \right) + r^2 \dot{\theta}^2 \right) - mg \frac{r}{\tan \alpha} = \\ &= \frac{m}{2} \left(\frac{\dot{r}^2}{\sin^2 \alpha} + r^2 \dot{\theta}^2 \right) - \frac{mgr}{\tan \alpha} \end{aligned}$$

$$\begin{aligned} \boxed{\theta} \quad \frac{\partial \mathcal{L}}{\partial \theta} &= 0 \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) &= \frac{d}{dt} (mr^2 \dot{\theta}) \Rightarrow \frac{d}{dt} (mr^2 \dot{\theta}) = 0 \Rightarrow mr^2 \dot{\theta} = L_0 \quad \text{constant} \\ &\Rightarrow \boxed{\dot{\theta} = \frac{L_0}{mr^2}} \end{aligned}$$

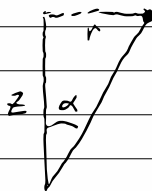
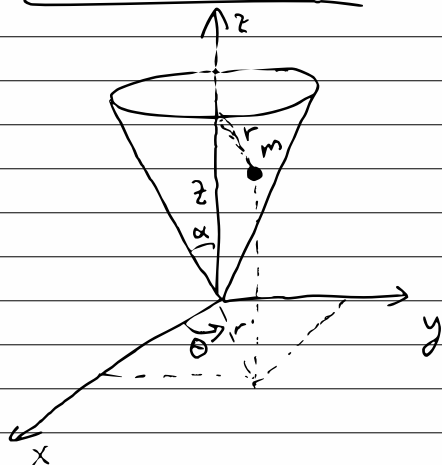
$$\boxed{r} \quad \frac{\partial \mathcal{L}}{\partial r} = mr\dot{\theta}^2 - \frac{mg}{\tan \alpha} = \frac{L_0^2}{mr^3} - \frac{mg}{\tan \alpha} \quad \Rightarrow$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) = \frac{d}{dt} \left(\frac{m\dot{r}}{\sin^2 \alpha} \right) = \frac{m}{\sin^2 \alpha} \ddot{r}$$

$$\Rightarrow \boxed{\ddot{r} = \sin^2 \alpha \left(\frac{L_0^2}{m^2 r^3} - \frac{g}{\tan \alpha} \right)}$$

Particle in a cone with forces of constraint

3.10



$$\dot{r} = \tan \alpha \dot{z}$$

The constraint is $\frac{r}{z} = \tan \alpha$

$$\begin{aligned} x &= r \cos \theta \Rightarrow \dot{x} = \dot{r} \cos \theta - r \sin \theta \dot{\theta} \Rightarrow \dot{x}^2 = \dot{r}^2 \cos^2 \theta + r^2 \sin^2 \theta \dot{\theta}^2 - 2r \dot{r} \cos \theta \sin \theta \dot{\theta} \\ y &= r \sin \theta \Rightarrow \dot{y} = \dot{r} \sin \theta + r \cos \theta \dot{\theta} \Rightarrow \dot{y}^2 = \dot{r}^2 \sin^2 \theta + r^2 \cos^2 \theta \dot{\theta}^2 + 2r \dot{r} \cos \theta \sin \theta \dot{\theta} \\ z &= z \Rightarrow \dot{z} = \dot{z} \Rightarrow \dot{z}^2 = \dot{z}^2 \end{aligned}$$

$$\dot{x}^2 + \dot{y}^2 + \dot{z}^2 = \dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2;$$

$$\eta = \frac{r}{z} - \tan \alpha; \quad \frac{\partial \eta}{\partial r} = \frac{1}{z}; \quad \frac{\partial \eta}{\partial z} = -\frac{r}{z^2}; \quad \frac{\partial \eta}{\partial \theta} = 0$$

$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) - mgz$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = \frac{d}{dt} (mr^2 \dot{\theta}) = 0 \Rightarrow \dot{\theta} = \frac{l_0}{mr^2}$$

$$z = \frac{r}{\tan \alpha}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) - \frac{\partial \mathcal{L}}{\partial r} = \lambda \frac{\partial \eta}{\partial r} \Rightarrow m \ddot{r} - m \dot{\theta}^2 r = \frac{\lambda}{z} \Rightarrow$$

$$\Rightarrow m \ddot{r} - m \frac{l_0^2}{m^2 r^4} r = \frac{\lambda \tan \alpha}{r} \Rightarrow \ddot{r} = \frac{l_0^2}{m^2 r^3} + \frac{\lambda \tan \alpha}{mr}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{z}} \right) - \frac{\partial \mathcal{L}}{\partial z} = \lambda \frac{\partial \eta}{\partial z} \Rightarrow m \ddot{z} - mg = -\frac{\lambda r}{z^2} \Rightarrow \begin{cases} z = \frac{r}{\tan \alpha} \\ \ddot{z} = -\frac{\ddot{r}}{\tan \alpha} \end{cases}$$

$$\Rightarrow m \frac{\ddot{r}}{\tan \alpha} + mg = -\frac{\lambda r}{r} \Rightarrow \text{We use this equation to get } \lambda$$

$$\Rightarrow \boxed{\lambda = -\frac{mr}{\tan \alpha} \left(\frac{\ddot{r}}{\tan \alpha} + g \right)}$$

$$\theta = \frac{l}{mr^2}$$

$$\ddot{r} = \frac{l \ddot{\theta}}{m^2 r^3} + \frac{\lambda \tan \alpha}{mr} = \frac{l \ddot{\theta}}{m^2 r^3} - \left(\frac{\ddot{r}}{\tan \alpha} + g \right) \frac{1}{\tan \alpha} \Rightarrow$$

$$\Rightarrow \ddot{r} \left(1 + \frac{1}{\tan^2 \alpha} \right) = \frac{l \ddot{\theta}}{m^2 r^3} - \frac{g}{\tan \alpha} \Rightarrow \boxed{\ddot{r} = \left(\frac{l \ddot{\theta}}{m^2 r^3} - \frac{g}{\tan \alpha} \right) \sin^2 \alpha}$$

$$1 + \frac{1}{\tan^2 \alpha} = \frac{1}{\sin^2 \alpha}$$

Now let's write λ without $\ddot{r}$ (forces don't depend on acceleration)

we have it in terms of θ

$$\lambda = -\frac{mr}{\tan \alpha} \left(\frac{\ddot{r}}{\tan \alpha} + g \right) = -\frac{mr}{\tan \alpha} \left[-\frac{r \ddot{\theta}}{\tan \alpha} \sin^2 \alpha - \frac{g}{\tan \alpha} \sin^2 \alpha + g \right]$$

$$= -\frac{mr}{\tan \alpha} \left[r \ddot{\theta} \sin \alpha \cos \alpha + g \sin^2 \alpha \right]$$

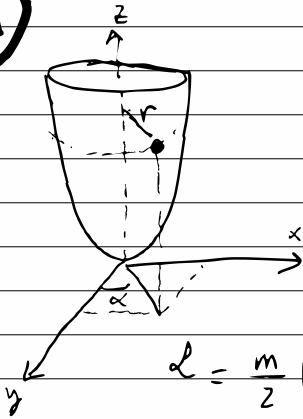
Now the forces of constraint are $\lambda \frac{\partial \eta}{\partial z}$ and $\lambda \frac{\partial \eta}{\partial r}$

$$\Rightarrow \boxed{F_z = \lambda \frac{\partial \eta}{\partial z} = \frac{r}{z^2} \frac{mr}{\tan \alpha} \left[r \ddot{\theta} \sin \alpha \cos \alpha + g \sin^2 \alpha \right] = m(r \ddot{\theta} \sin \alpha \cos \alpha + g \sin^2 \alpha)}$$

$$\boxed{F_r = \lambda \frac{\partial \eta}{\partial r} = \frac{1}{2} \frac{mr}{\tan \alpha} \left[r \ddot{\theta} \sin \alpha \cos \alpha + g \sin^2 \alpha \right] = -m(r \ddot{\theta} \cos^2 \alpha + g \sin \alpha \cos \alpha)}$$

Particle inside paraboloid

3.11



$$x^2 + y^2 = Rz \Rightarrow \text{Polar: } r^2 = Rz$$

$$x = r \sin \alpha; \dot{x} = \dot{r} \sin \alpha + r \cos \alpha \dot{\alpha}$$

$$y = r \cos \alpha; \dot{y} = \dot{r} \cos \alpha - r \sin \alpha \dot{\alpha}$$

$$z = \frac{r^2}{R}; \dot{z} = \frac{2r\dot{r}}{R}$$

$$(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = \dot{r}^2 + r^2 \dot{\alpha}^2 + \frac{4r^2 \dot{r}^2}{R^2}$$

$$\mathcal{L} = \frac{m}{2} \left(\dot{r}^2 + r^2 \dot{\alpha}^2 + \frac{4r^2 \dot{r}^2}{R^2} \right) - mgz =$$

$$= \frac{m}{2} \left(1 + \frac{4r^2}{R^2} \right) \dot{r}^2 + \frac{mr^2}{2} \dot{\alpha}^2 - mg \frac{r^2}{R}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\alpha}} \right) = 0 \Rightarrow \text{ct.} = \ell = \frac{\partial \mathcal{L}}{\partial \dot{\alpha}} = mr^2 \dot{\alpha} \Rightarrow \boxed{\dot{\alpha} = \frac{\ell}{mr^2}}$$

$$\frac{\partial \mathcal{L}}{\partial r} = \frac{m}{2} \frac{8r}{R^2} \dot{r}^2 + mr \frac{\ell^2}{m^2 r^4} - \frac{2m\dot{r}}{R} = \frac{4mr}{R^2} \dot{r}^2 + \frac{\ell^2}{mr^3} - \frac{2m\dot{r}}{R}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) = \frac{d}{dt} \left(m \left(1 + \frac{4r^2}{R^2} \right) \dot{r} \right) = m \left(1 + \frac{4r^2}{R^2} \right) \ddot{r} + \frac{m 8r \dot{r}^2}{R^2}$$

$$\Rightarrow m \left(1 + \frac{4r^2}{R^2} \right) \ddot{r} + \frac{8mr \dot{r}^2}{R^2} = \frac{4mr}{R^2} \dot{r}^2 + m r \dot{\alpha}^2 - \frac{2m\dot{r}}{R}$$

$$\Rightarrow \boxed{\left(1 + \frac{4r^2}{R^2} \right) \ddot{r} = - \frac{4r}{R^2} \dot{r}^2 + r \dot{\alpha}^2 - \frac{2\dot{r}}{R}}$$

(b) Prove that particle will stay at $r = r_0 = \text{const}$ if

$$\omega = \sqrt{\frac{2g}{R}} = \dot{\alpha}$$

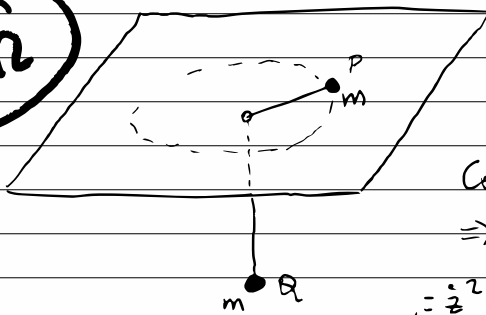
$$\left(1 + \frac{4r^2}{R^2}\right) \ddot{r} = - \frac{4r}{R^2} \dot{r}^2 + r \dot{\alpha}^2 - \frac{2gr}{R}$$

$$\left(\frac{1 + 4r^2}{R^2}\right) \ddot{r} + \frac{4r}{R^2} \dot{r}^2 = \frac{r}{R} \dot{\alpha}^2 - \frac{2gr}{R} = 0$$

$$\Rightarrow \ddot{r} + \frac{4r^2 \ddot{r}}{R^2} + \frac{4r \dot{r}^2}{R^2} = 0 \quad \text{and } r = ct \text{ is obviously a solution.}$$

$$(c) \dot{\alpha} = \frac{l}{mr_0^2} = \sqrt{\frac{2g}{R}} \Rightarrow l^2 = \frac{2mg r_0^4}{R}$$

3.12



P $\rightarrow$ polar coordinates r, θ
Q $\rightarrow$ z coordinate alone

Constraint $z + r = L = \text{const.}$

$$\Rightarrow z = L - r$$

$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) + mg(L - r)$$

$\downarrow = \dot{z}^2$

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0 \Rightarrow \text{const.} \equiv l_0 = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \dot{\theta} \Rightarrow \boxed{\dot{\theta} = \frac{l_0}{m r^2}}$$

$$\left. \begin{aligned} \frac{\partial \mathcal{L}}{\partial r} &= m r \dot{\theta}^2 - mg \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) &= 2 m \ddot{r} \end{aligned} \right\} \Rightarrow \boxed{\ddot{r} = \frac{r \dot{\theta}^2}{2} - \frac{g}{2}} = \boxed{\frac{l_0^2}{2 m^2 r^3} - \frac{g}{2}}$$

(b) Circular motion?

$$\ddot{r}|_{r_0} = 0 \Rightarrow \frac{r_0}{2} \frac{l_0^2}{m^2 r_0^3} = \frac{g}{2} \Rightarrow \boxed{r_0^3 = \frac{l_0^2}{m^2 g}}$$

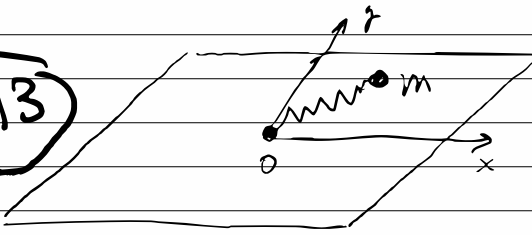
(c) $r = r_0 + \delta \Rightarrow \ddot{r} = \ddot{\delta}$

$$\frac{1}{r^3} = \frac{1}{r_0^3} \frac{1}{(1 + \frac{\delta}{r_0})^3} \approx \frac{1}{r_0^3} \left(1 - \frac{3\delta}{r_0} \right)$$

$$\Rightarrow \ddot{\delta} = \frac{l_0^2}{2 m^2 r_0^3} \left(1 - \frac{3\delta}{r_0} \right) - \frac{g}{2} \Rightarrow$$

$$\Rightarrow \ddot{\delta} = - \frac{3}{2} \frac{l_0^2}{m^2 r_0^4} \delta = - \frac{3}{2} \frac{g}{r_0} \delta$$

3.13



$$a) \mathcal{L} = \frac{m}{2} (\dot{x}^2 + \dot{y}^2) - \frac{k}{2} (x^2 + y^2)$$

$$b) x \rightarrow x + \epsilon y; \quad y \rightarrow y - \epsilon x$$

$$\left\{ \begin{array}{l} \dot{x} \rightarrow \dot{x} + \epsilon \dot{y}; \quad \dot{y} \rightarrow \dot{y} - \epsilon \dot{x} \\ \dot{x}^2 \rightarrow \dot{x}^2 + 2\dot{x}\dot{y}\epsilon + \epsilon^2 \dot{y}^2 \\ \dot{y}^2 \rightarrow \dot{y}^2 - 2\dot{x}\dot{y}\epsilon + \epsilon^2 \dot{x}^2 \end{array} \right\} \quad \begin{array}{l} x^2 \rightarrow x^2 + \epsilon^2 y^2 + 2xy\epsilon \\ y^2 \rightarrow y^2 + \epsilon^2 x^2 - 2xy\epsilon \end{array}$$

$$\begin{aligned} \mathcal{L} &\rightarrow \frac{m}{2} (\dot{x}^2 + 2\dot{x}\dot{y}\epsilon + \epsilon^2 \dot{y}^2 + \dot{y}^2 - 2\dot{x}\dot{y}\epsilon + \epsilon^2 \dot{x}^2) \\ &\quad - \frac{k}{2} (x^2 + \epsilon^2 y^2 + 2xy\epsilon + y^2 + \epsilon^2 x^2 - 2xy\epsilon) = \\ &= \frac{m}{2} (\dot{x}^2 + \dot{y}^2) - \frac{k}{2} (x^2 + y^2) + \mathcal{O}(\epsilon^2) \quad \square. \end{aligned}$$

$$c) p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x}; \quad p_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = m \dot{y}$$

Noether's Theorem: $\sum_i p_i K_i$ is constant

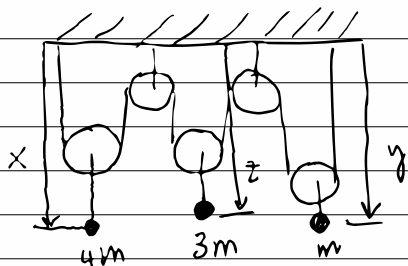
Our transformations are $x \rightarrow x + \epsilon y$ and $y \rightarrow y - \epsilon x$

so $K_x = y$ and $K_y = -x$

$$\Rightarrow m \dot{x} y - m \dot{y} x = \text{constant} \Rightarrow m(\dot{x} y - x \dot{y}) = L_z$$

L_z is the angular momentum in z .

3.14



Constraint

If the masses at the ends go up by x and y , then the central mass goes down $x+y$!

$$\Rightarrow z = (L - x - y) \Rightarrow \dot{z} = -\dot{x} - \dot{y}$$

$$\mathcal{L} = \frac{4m}{2} \dot{x}^2 + \frac{m}{2} \dot{y}^2 + \frac{3m}{2} (\dot{x} + \dot{y})^2 + 4mgx + 3mg(L - x - y) + mgy$$

$$= \frac{m}{2} (4\dot{x}^2 + \dot{y}^2 + 3\dot{x}^2 + 3\dot{y}^2 + 6\dot{x}\dot{y}) +$$

$$+ mg(4x + 3L - 3x - 3y + y) =$$

$$= \frac{m}{2} (7\dot{x}^2 + 4\dot{y}^2 + 6\dot{x}\dot{y}) + 3mgL + mg(x - 2y)$$

Symmetry: $x \rightarrow x + 2\epsilon \Rightarrow K_x = 2$
 $y \rightarrow y + \epsilon \Rightarrow K_y = 1$

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = 7m\dot{x} + 3m\dot{y}$$

$$p_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = 4m\dot{y} + 3m\dot{x}$$

$$\Rightarrow \boxed{\mathcal{P} = 14m\dot{x} + 6m\dot{y} + 4m\dot{y} + 3m\dot{x} = m(17\dot{x} + 10\dot{y})}$$